Face Recognition under Challenging Image Conditions

AI Developer Intern technical assignment · 23 October 2026

1. Introduction

Objective

Recognise eight people from the company-provided images, many of which are small, blurred, dark or seen from the side. A pre-trained face-embedding network is fine-tuned on a training split of those images, and the effect is measured against the unmodified network on a held-out 20% test split.

Problem definition

Two tasks are measured separately. **Verification**: given two face images, decide whether they show the same person by thresholding the cosine similarity of their embeddings. **Identification**: given one face, name which of the eight enrolled people it is, using a k-nearest-neighbour classifier over embeddings of the training images. Verification is the primary comparison between the baseline and the fine-tuned model because it isolates the quality of the embedding itself.

Selected architecture

Faces are located with a YOLOv8n detector trained for faces and cropped to a 112 x 112 square. Embeddings come from InsightFace's ArcFace IResNet-50 (*w600k_r50*, trained on WebFace600K), which produces a 512-dimensional vector. InsightFace distributes this model only as an inference ONNX file, so it was rebuilt as a PyTorch module and its weights copied across. The copy reproduces the ONNX embeddings to a cosine similarity above 0.99999, which makes the baseline genuinely the released model and the fine-tuned model the same architecture with updated weights. Fine-tuning updated 25,962,496 of 43,578,176 parameters.

Component	Technology
Language	Python 3.11
Face detection	Ultralytics YOLOv8n, lindevs yolov8n-face (WIDER FACE), release 1.0.1
Face embedding	InsightFace ArcFace IResNet-50 (buffalo_l / w600k_r50), PyTorch re-implementation
Training	PyTorch 2.12.0+cu130, ArcFace margin loss, mixed precision on GPU
Identity classifier	scikit-learn KNeighborsClassifier, cosine distance
Image processing	OpenCV, NumPy
Data and evaluation	pandas, scikit-learn metrics
Visualisation and report	Matplotlib, ReportLab
Logging	structlog
Hardware	NVIDIA GeForce RTX 2050

Table 1. *Technology stack.*

2. Dataset and Preprocessing

Dataset summary

The archive holds 22,286 files for eight people. Each person has a *high_quality* folder of frames from one close-up video and a *low_quality* folder of small CCTV-style crops. All images are already face crops. After exclusions, 19,846 images were used.

Identity	High quality	Low quality	Train	Validation	Test
ananthu	709	3492	2834	489	878
firoz	593	862	933	172	350
mujeeb	255	843	692	104	302
ruvais	104	976	745	111	224
sebin	369	1618	1305	244	438
shinjil	254	1488	1184	184	374
suresh	579	6331	4505	829	1576
thomas	554	819	945	152	276
Total	3417	16429	13143	2285	4418

Table 2. *Images used per identity, by source quality and split subset.*

Image quality challenges

High-quality crops are around 300-500 pixels wide. Low-quality crops are mostly 16-80 pixels, frequently in profile or looking down, motion-blurred, compressed, or partly showing the back of the head. Consecutive video frames are nearly identical, so a naive file-level split would put near-copies of test images in training.

Face detection

YOLOv8n-face runs on each image after small crops are upscaled to at least 160 pixels and padded, because the detector was trained on whole scenes rather than tight crops. The most confident box is squared and resized to 112 x 112. When no face is found, the whole crop is used instead (the images are already face crops). This fallback is recorded for every image and reported below. Crop margin 0.0 was chosen on validation pairs with the pre-trained model.

Detection status	Train	Validation	Test
full image fallback	7220	1224	2410
detected	5923	1061	2008

Table 3. *Detection outcome per subset. Fallback images are still evaluated and are counted in every metric.*

Training augmentation

Augmentation	Method
none	Face crop as it is
bright	Linear gain 1.2 and offset +35
dark	Gamma 1.9
contrast	CLAHE on the LAB lightness channel
grayscale	Luminance copied to all three channels
shadow	Linear shading ramp at a random angle
flip	Horizontal mirror, applied to half of all samples on top of the above

Table 4. *Augmentations applied in memory to training face crops. One is drawn at random for every sample; no augmented images are written to disk, and validation and test images are never augmented.*

80/20 split strategy

Each sequence (identity, quality and file type) was cut into contiguous segments of at most 50 frames, with at least 10 segments per sequence. Whole segments were assigned to test until about 20.0% of each sequence was held out (20.1% overall), and 15.0% of the remaining segments became a training-only validation set. Seed

42. Exclusions: 154 unreadable files, 178 exact duplicates (first copy kept), 1602 training images that were near-copies of a test image, and 506 that were near-copies of a validation image (thumbnail cosine at least 0.95). Because only training images were removed, the test set is 22.3% of the images finally used. The split is copied to data/train/<person>/high_quality|low_quality and data/test/<person>/high_quality|low_quality, the same layout as the original archive; validation images stay in data/train and are marked in the training manifest.

Dataset integrity checks

Before training and before each evaluation, the code verifies that no source ID, file, SHA-256 hash or segment appears in both partitions and that validation and training segments are disjoint; any overlap stops the run. The full audit of every artefact (near-duplicates recomputed from raw files, checkpoints, KNN gallery, pair files) is in Section 6.

3. Baseline Evaluation

The unmodified pre-trained network was evaluated before any fine-tuning. For each identity and each quality combination (high-high, high-low, low-low), same-person pairs were drawn from different segments and matched with an equal number of different-person pairs. The pair list is saved once and reused unchanged for the fine-tuned model. The decision threshold maximises accuracy on a separate set of validation pairs; test labels were not used to choose it.

Metric	Baseline
ROC-AUC	0.7108
Average precision	0.7313
Verification accuracy	0.6537
Precision	0.6843
Recall	0.5712
F1-score	0.6226
False acceptance rate	0.2636
False rejection rate	0.4288
Threshold (from validation pairs)	0.1585
Test pairs	7,922

Table 5. *Baseline verification metrics on the held-out test pairs.*

Pair type	Pairs	ROC-AUC	Accuracy
hq-hq	2,611	0.9218	0.8598
hq-lq	2,655	0.6127	0.5522
lq-lq	2,656	0.5903	0.5527

Table 6. *Baseline verification by image-quality combination.*

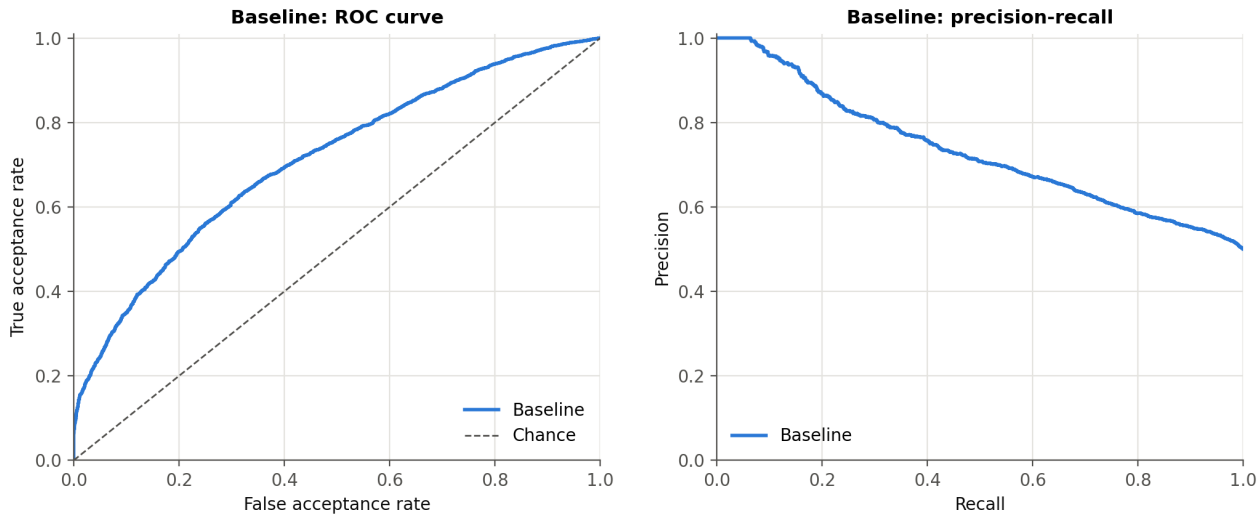


Figure 1. *Baseline ROC curve (left) and precision-recall curve (right) on the test pairs.*

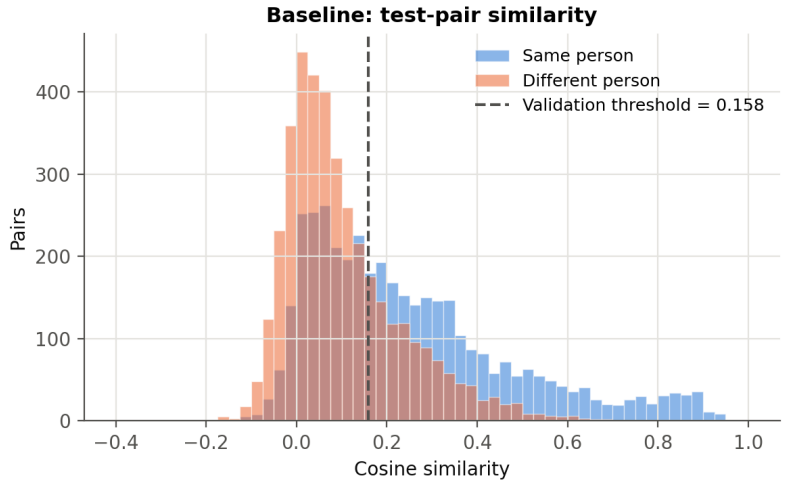


Figure 2. *Baseline cosine similarity of same-person and different-person test pairs, with the validation threshold.*

4. Fine-Tuning and Training

The backbone was fine-tuned with an ArcFace additive-angular-margin softmax over the eight identities. Triplet loss was not used: with eight identities, almost every batch would contain only easy negatives. The classification head was started from each identity's mean pre-trained embedding, so early gradients do not disturb the backbone. The first three stages of the network were frozen and BatchNorm running statistics were held at their pre-trained values. Each epoch sampled identities equally often and, for each sampled image, one random augmentation from Table 4 plus a horizontal flip half of the time.

Setting	Value
Architecture	IResNet-50, 512-d embedding, 112 x 112 RGB input, (x - 127.5) / 127.5
Pre-trained weights	InsightFace buffalo_l w600k_r50.onnx (WebFace600K, ArcFace)
Trainable modules	layer4, bn2, fc, features (25,962,496 parameters)
Frozen modules	conv1, prelu, layer1-layer3 (17,615,680 parameters)
Loss	ArcFace margin softmax, scale 32.0, margin 0.3
Optimizer	adamw, lr 0.0001 backbone / 0.001 head, weight decay 0.0005
Scheduler	cosine over 10 epochs, early stopping patience 3
Batch size	64
Sampling	identity-balanced weighted sampling with replacement
Augmentation	none, bright, dark, contrast, grayscale, shadow; horizontal flip (in memory)
Training data	13,143 training-subset images
Validation	2,285 validation images from held-out training segments
Checkpoint selection	best validation-pair ROC-AUC (epoch 0 = pre-trained model)
Hardware	NVIDIA GeForce RTX 2050, PyTorch 2.12.0+cu130
Epochs run / best epoch	10 / 8
Measured training time	16.3 minutes
Random seed	42

Table 7. *Fine-tuning configuration as recorded during the run.*

Epoch	Train loss	Train acc.	Val. loss	Val. acc.	Val. ROC-AUC	Seconds
0	-	-	7.803	0.712	0.6586	0
1	3.105	0.810	2.835	0.819	0.9119	102
2	1.334	0.922	1.948	0.882	0.9518	77
3	0.919	0.949	1.844	0.891	0.9589	88
4	0.657	0.963	1.676	0.904	0.9644	87
5	0.444	0.977	1.703	0.904	0.9554	81
6	0.335	0.983	1.523	0.912	0.9620	83
7	0.230	0.988	1.519	0.918	0.9663	94
8	0.172	0.992	1.527	0.912	0.9700	92
9	0.147	0.993	1.399	0.926	0.9675	86
10	0.117	0.994	1.398	0.926	0.9674	89

Table 8. *Training history. Accuracies are head predictions without the margin.*

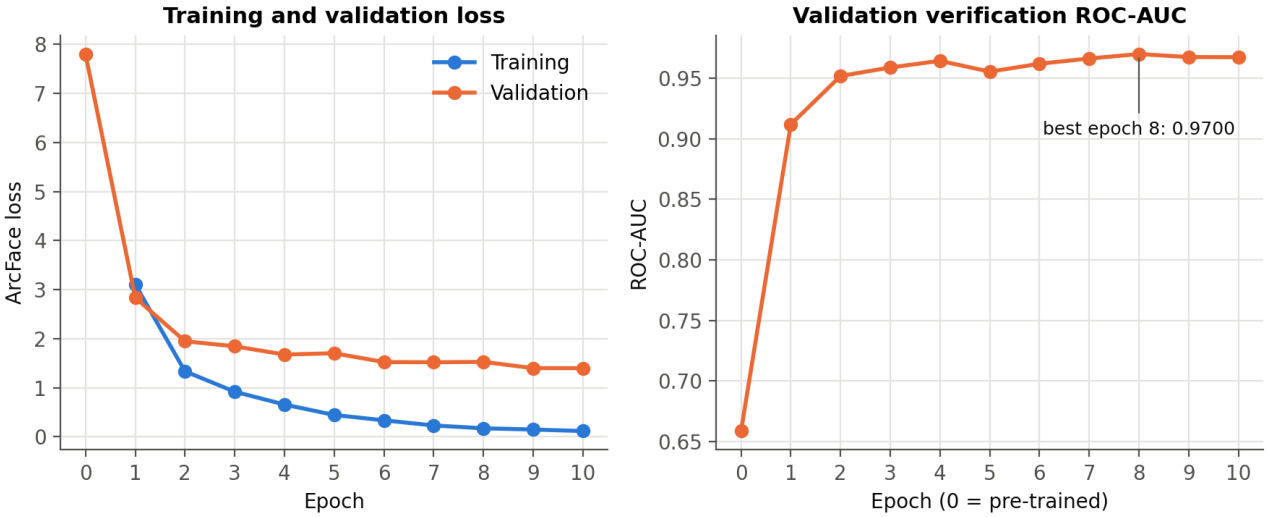


Figure 3. *Training and validation loss (left) and validation-pair ROC-AUC (right) per epoch.*

To confirm that fine-tuning updated the network, every module of best_model.pt was compared with the pre-trained weights. Modules with a non-zero relative change: layer4 (0.1011), bn2 (0.0901), fc (0.6540), features (0.0043). All frozen modules are unchanged.

5. Final Evaluation and Comparison

The fine-tuned model was evaluated with the baseline's exact protocol: the same cached detections, the same test pairs (identical SHA-256), a threshold calibrated on the same validation pairs, and KNN neighbour count chosen on validation accuracy. The two runs differ only in the embedding weights.

Metric	Baseline	Fine-tuned	Change	Direction
ROC-AUC	0.7108	0.9732	+0.2624	improved
Average precision	0.7313	0.9753	+0.2440	improved
Verification accuracy	0.6537	0.9149	+0.2612	improved
Precision	0.6843	0.9480	+0.2637	improved
Recall	0.5712	0.8781	+0.3069	improved
F1-score	0.6226	0.9117	+0.2890	improved
False acceptance rate	0.2636	0.0482	-0.2154	improved
False rejection rate	0.4288	0.1219	-0.3069	improved
KNN identification accuracy	0.8345	0.9140	+0.0794	improved
KNN macro F1	0.7826	0.8848	+0.1022	improved

Table 9. *Test-set metrics before and after fine-tuning. False acceptance and rejection rates are better when lower.*

Pair type	Baseline ROC-AUC	Fine-tuned ROC-AUC	Baseline accuracy	Fine-tuned accuracy
hq-hq	0.9218	0.9995	0.8598	0.9851
hq-lq	0.6127	0.9647	0.5522	0.8960
lq-lq	0.5903	0.9392	0.5527	0.8648

Table 10. *Verification by image-quality combination.*

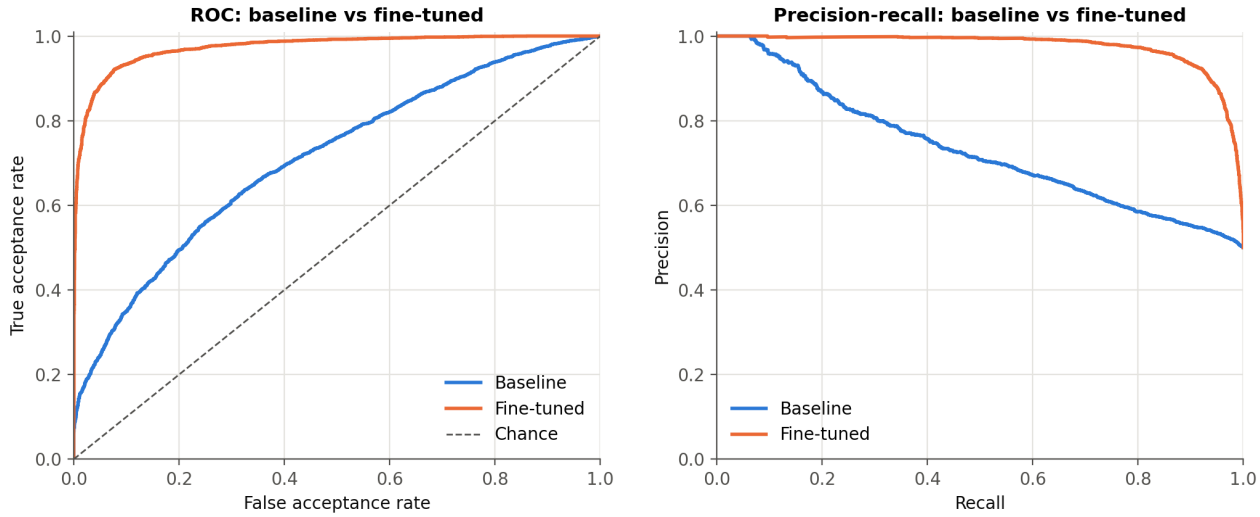


Figure 4. *ROC (left) and precision-recall (right) curves of both models on the same test pairs.*

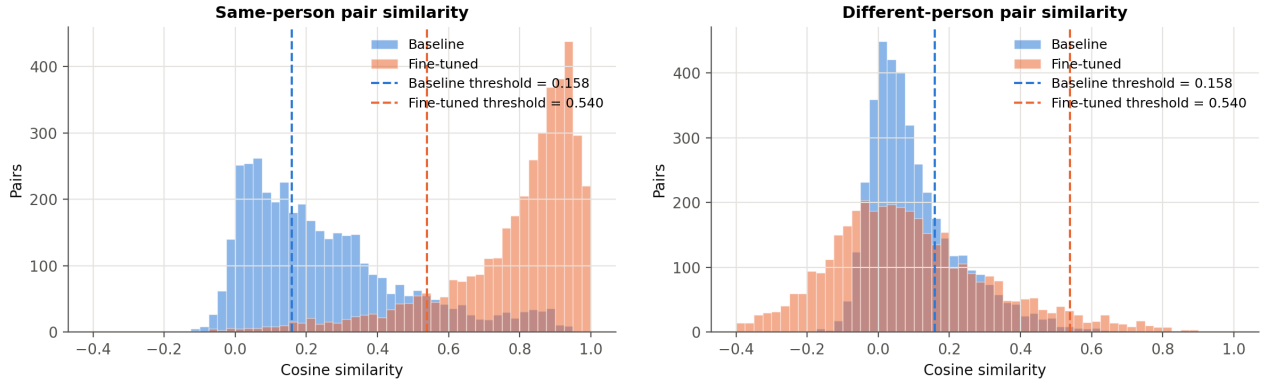


Figure 5. *Similarity of same-person pairs (left) and different-person pairs (right) for both models, with each model's validation threshold.*

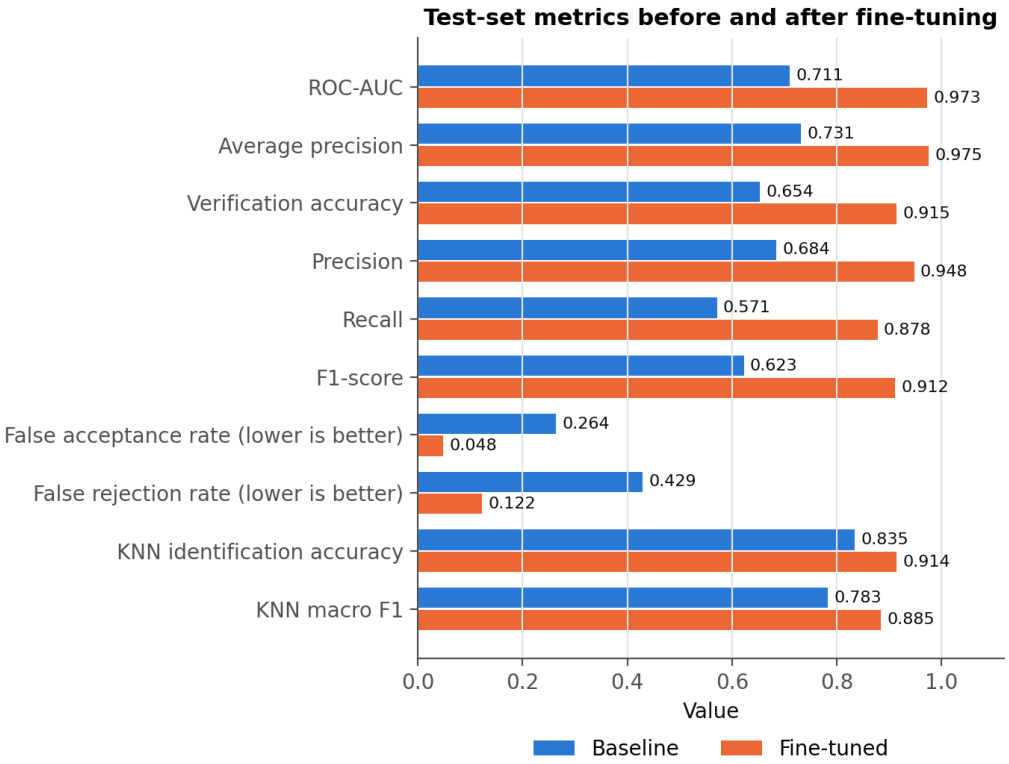


Figure 6. *Before-and-after comparison of test-set metrics.*

Identification with the saved KNN classifier

Part 3 of the notebook loaded best_model.pt and knn_classifier.joblib from disk (k = 15) and classified all 4,418 test images; 0 had no usable face and are counted as errors. Closed-set accuracy was 0.9140 (macro F1 0.8848): 0.9907 on high-quality and 0.8954 on low-quality images. With the validation-calibrated unknown-identity threshold (0.5398), 0 images were rejected as unknown. Since every test person is enrolled, each rejection counts as an error, giving accuracy 0.9140; accuracy on the accepted faces was 0.9140. For comparison, the baseline embeddings with the same KNN procedure reached 0.8345 closed-set accuracy.

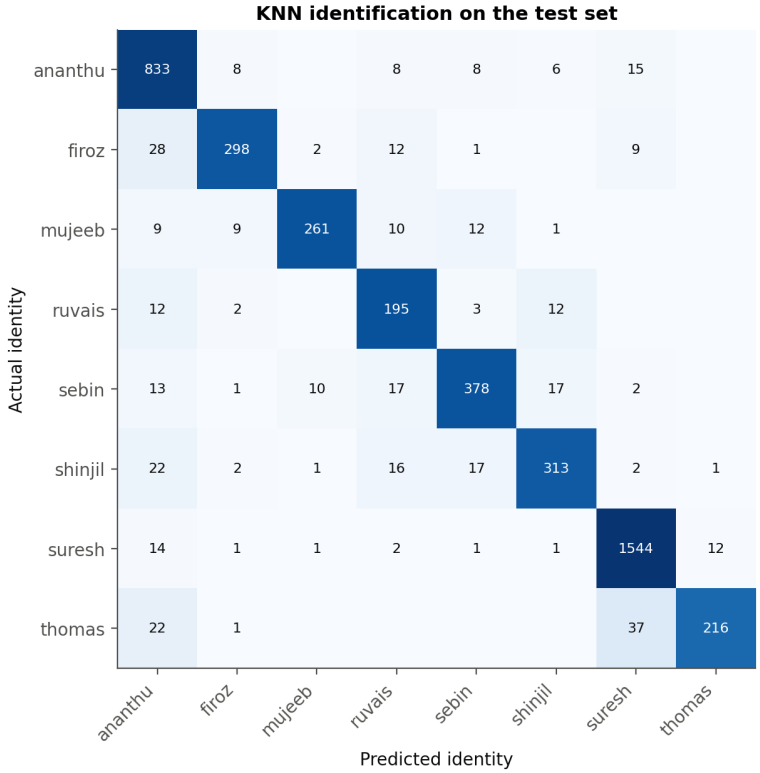


Figure 7. *Confusion matrix of the saved KNN classifier on the test set (closed-set predictions, counts).*

Improvements and regressions

- Improved: ROC-AUC, Average precision, Verification accuracy, Precision, Recall, F1-score, False acceptance rate, False rejection rate, KNN identification accuracy, KNN macro F1.
- Regressed: none.

Failure analysis

The weakest identity and quality groups by closed-set accuracy were thomas low-quality (0.623), firoz low-quality (0.740), mujeeb low-quality (0.795), shinjil low-quality (0.809). The most frequent confusions were thomas predicted as suresh (37); firoz predicted as ananthu (28); shinjil predicted as ananthu (22); thomas predicted as ananthu (22); sebin predicted as ruvais (17). By detection outcome, accuracy was detected 0.961 (2,008 images), full image fallback 0.875 (2,410 images). Low-quality CCTV crops dominate the errors: many are under 32 pixels wide, seen in profile or from behind, and too small for the face detector.

For low-quality to low-quality pairs, ROC-AUC moved from 0.5903 to 0.9392.

6. Integrity, Reproducibility and Conclusion

Data leakage checks

Output of scripts/check_data_leakage.py, which re-reads every artefact independently of the training code.

Check	Status	Detail
Train/test manifests are disjoint	passed	no shared source IDs, files, SHA-256 hashes or segments (6 overlap counts are all 0)
No near-duplicates across train/test	passed	highest train-to-test thumbnail similarity 0.94999; 0 training images at or above 0.95
data/train and data/test share no files	passed	15428 train and 4418 test files, no file content in both
best_model.pt trained without test data	passed	fine-tuned on 13143 training-subset images (epoch 8); no validation or test images
final_model.pt trained without test data	passed	fine-tuned on 13143 training-subset images (epoch 10); no validation or test images
KNN gallery and threshold exclude test data	passed	gallery of 13143 training images; threshold from validation pairs; matches best_model.pt
Pair manifests stay within their subsets	passed	7922 test pairs use only test images; 3911 threshold-calibration pairs use only validation images
Baseline and final use one fixed protocol	passed	2 evaluation(s) used the same test pairs and validation-calibrated thresholds

Table 11. *Integrity audit.*

Reproducibility

- Seed 42 for the split, pair sampling, data sampler, augmentation and KNN selection; all settings in src/config.py and stored in each checkpoint.
- Manifests with SHA-256 hashes for every image: data/manifests/train_manifest.csv, test_manifest.csv, excluded_manifest.csv.
- Fixed test pairs (SHA-256 183fa05eba9e4125...) shared by both evaluations.
- Run order: notebook Part 1, scripts.baseline_evaluation, notebook Parts 2-3, scripts.final_evaluation, scripts.check_data_leakage, scripts.generate_report. Commands are listed in README.md.
- Hardware: NVIDIA GeForce RTX 2050; GPU kernels are not forced to be deterministic, so retraining may differ in late decimals.

Limitations

- Each person's high-quality images come from essentially one recording. Segment splitting and near-duplicate removal stop near-identical frames crossing the split, but training and test frames still share lighting, clothing and background.
- Training and testing use the same eight people, so the results show how well the model separates these people. They do not show generalisation to new identities, and the fine-tuned embedding may be worse than the original on unseen faces.
- The face detector returns boxes without landmarks, so faces are box-cropped rather than landmark-aligned as ArcFace expects, which lowers both models' scores.
- Low-quality crops in which the detector finds no face are used whole, so their framing differs from detected faces; accuracy by detection outcome is reported in Section 5.
- Near-duplicate detection compares 32 x 32 thumbnails. It catches near-identical frames, not every frame from the same moment under a different crop.
- The unknown-identity threshold is the validation verification threshold applied to the nearest gallery face. The data contains no unenrolled people, so open-set rejection could not be evaluated; on the closed test set it rejected none of the images.

- One training run was made. No variance across seeds was measured.

Conclusion

On the fixed test pairs, verification ROC-AUC changed from 0.7108 to 0.9732, and verification accuracy at the validation-calibrated threshold from 0.6537 to 0.9149. KNN identification accuracy under the same protocol changed from 0.8345 to 0.9140. Given the limitations above, these figures measure recognition of the eight enrolled people under the conditions in the provided images.

Declaration

I declare that the 20% evaluation subset described in this report was not used for training, fine-tuning, checkpoint selection, KNN hyperparameter selection or threshold calibration, that only the company-provided images were used for the reported experiments, and that all results were produced by the code submitted with this report.

Name: _____

Signature: _____

Date: _____